# Homework 4

Due Wednesday, 23 September 2026

1. Let $X_1, \ldots, X_n$ be a random sample from the pdf

$$f(x;\theta) = \theta x^{-2}, \qquad 0 < \theta \leq x < \infty.$$

   a. Find the method of moments estimator of $\theta$. (hint: don't overthink this).
   b. Find the maximum likelihood estimator of $\theta$.

2. Let $X_1, \ldots, X_n$ be iid with pdf

$$f(x;\theta) = \frac{1}{\theta} \quad 0 \leq x \leq \theta; \theta > 0$$

   Estimate $\theta$ using both the method of moments and maximum likelihood. Calculate the means and variances of the two estimators. Which one should be preferred and why?

3. Suppose that $X_1, \ldots, X_n$ are a random sample from a Uniform$(0, 2\theta + 1)$ distribution.

   a. Find the MLE of $\theta$.
   b. Obtain the MLE of the **variance** of the underlying Uniform$(0, 2\theta + 1)$ distribution.

4. Let $Y_1, \ldots, Y_n$ be iid random variables from the population with pdf

$$f(y;\beta) = \begin{cases} 3\beta^3 y^{-4} & y \geq \beta \\ 0 & \text{elsewhere} \end{cases}$$

   Find the MLE of $\beta$.

5. Suppose the random variables $Y_1, \ldots, Y_n$ satisfy

$$Y_i = \beta x_i + e_i, \qquad i = 1, \ldots, n$$

   where $x_1, \ldots, x_n$ are fixed constants and $e_1, \ldots, e_n$ are iid N$(0, \sigma^2)$ random variables, with $\sigma^2$ unknown.

   a. Find the MLE of $\beta$, and find E$(\hat{\beta}_{\text{MLE}})$.
   b. Find the distribution (including parameters) of $\hat{\beta}_{\text{MLE}}$.